Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Homogeneous functions and Euler's theorem

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Homogeneous functions and Euler's theorem

1 Homogeneous functions and Euler's theorem

Study focus: Homogeneous functions and Euler's theorem. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll be learning something called as Euler's theorem, which is a wonderful observation made by Euler for a certain class of functions.

Key details

- The advantage and what are the uses for this Euler's theorem we can discuss later.
- Before we go to Euler's theorem, we need to refresh our memory about homogeneous functions.
- you might have heard of homogeneous function when you solved differential equations.
- And that was the most easiest one because any homogeneous function you could reduce into a variable separable form, right?
- I hope you remember this, right?
- So that means I was in a comfortable position to reduce this to variable separable by making the substitution y is equal to bx.
- But now what happens is we will discuss.
- We will see how to check a homogeneous function.
- I have a function x square plus y square.
- Or I may take a function which looks like x cube plus three x square y plus y cube plus three x y square.
- I know that this is homogeneous, but I'm not sure whether e whether e po of that is homogeneous.
- I don't know how to check this function to be homogeneous.
- If at all it is homogeneous, what is the degree of that function?
- One is the function homogeneous?
- And if it is homogeneous, what is its degree?

- Again, I'm restricting myself to a bivariate in the beginning.
- But if you want to go for a multivariable function, definitely you can do it by extending the definition.
- If it can be expressed in the form of either x to the power of n into some function of y by x, or if I can write it as y to the power of some m into some other function of x by y.
- Then we say that the function is homogeneous of degree n in this case and m in this case.
- What do I mean by that?
- How do I check if this is homogeneous?
- If I take out x square common, can I write this function as 1 plus y square pi x square.
- That means this function is homogeneous of degree.
- A function f of x y is said to be homogeneous of degree n.
- If it can be expressed in the form of x to the power of n into function of y by x or y par n into function of x by y.
- Whichever is convenient, you can choose according to the problem.
- Then how am I going to find if it is homogeneous or not?
- Again we'll use the same definition, but now it's not just y by x, but it is z by x also.
- you have x q plus y q plus z q plus three x y z I can take out x cube common and if I take out x cube common I can g get the remainder remaining terms in functions of y by x as well as z by x.

Example or application from the lecture

- But let me give you another example.
- Suppose I'll write a function which looks like sine inverse of x plus y plus z divided by something like x cube plus y cube plus z cube, or it could be tan inverse of something or e to the power of something.
- Let's say the definition says that a function u is equal to f of x y.

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Core explanation · 05:00–09:59

Or it could be other way also.

Key details

- It could be using z power n.
- Z power n into function of x by z and y by z.
- We call that is a homogeneous function of degree n negative.
- Or you can write this as y to the power of n into some other function of now x by y and z by y and w by y and that continues.
- So before I go to Euler's theorem,, by the way, he is Euler, as you can see in all his photographs, his one eye is shorter because he was blind by one eye.
- , in spite of that, he was an extraordinarily intelligent mathematician.
- He didn't had any former schooling or anything, but of course he was with the family of Legrange or Barnooli, I don't remember, but.

- That's how if you're interested you can read more about who Euler was and what all his contributions are.
- As you can see, a constant function definitely is a homogeneous function of degree zero.
- And of course the converse also works.
- And it need not be just positive integer.
- Degree can be positive, negative, zero, could be anything.
- And I can take out an x common from the denominator, right?
- This is 1 plus y by x.
- That means this function here is a homogeneous function of degree minus half.
- It's not always positive, negative.
- It could be rational, it could be positive, negative integer, or even zero.
- It's okay.
- Ah so the remaining results I'll come back to these things when I define Euler's theorem.
- This is the most important one.
- Then when I multiply x with the partial derivative of u with respect to x, y with the partial derivative of u with respect to y, it is same as n multiplied by the function itself.
- That's what I said.
- Check the degree, then you can say directly that x u x plus y u y.
- No wow, what is it happening geometrically?
- I'm finding the slope of the tangent parallel to x axis, but I'm multiplying that with x.
- I'm finding the slope or change along the y direction multiplied by y.

Example or application from the lecture

- Let's say you had another variable w, it should be w by x.
- And linear functions as we saw in the previous example, so we took 3x plus 4y and we saw that this linear function is of degree 1 always.
- So just to show you an example, suppose I had say root x plus root y divided by x plus y, if suppose this was my function, that I can take out root x common from the numerator and write this as 1 plus root of y by x.

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Core explanation · 10:00–14:59

There is a change that is happening along the y direction for a surface, right?

Key details

- You study all those changes, then what you get is stacking of the functions one above the other end times.
- That's what you get.
- Ux is 2x and uy will be 2y again.
- Plus 2 y square, or this is just 2 into the function itself.
- And this is a homogeneous function of degree 2.

- So basically what Euler observed is for a homogeneous function, homogeneous means which is smooth everywhere.
- For a homogeneous function,.
- The change along the x direction when scaled with x.
- Change along the y direction scaled with y will give you the entire function itself scaled by the factor of the degree.
- This is for the first derivative.
- We discussed this in the previous lectures.
- You have ux y and uyx.
- But definitely they are equal.
- Okay, because first order partial derivatives are continuous.
- So what Euler says is because this is second order, you scale it by 2.
- How many third order partial derivatives can exist?
- It is U XXX, right?
- Uy and then U XY.
- U X Y X.
- And then U X Y.
- And then U Y X and then U Y X Y.
- It's of order three, right?
- No x y x I did.
- X Y is also done.
- But anyway, so you can check the mixed order partial derivative and what happens to the right hand side will be n into n minus one into n minus two into u.
- That's what you're going to get.
- As in when you add.
- one more variable to it.
- Then if it is homogeneous, so u is equal to f of x y z is a homogeneous function of degree n, then I get x u x plus y u y plus z u z equal to u.
- At this point I want you to explore.
- What is the corollary one that we did?

Example or application from the lecture

- but let's see what is the extension of Euler's theorem.
- Suppose because I have only ux and u y.

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Core explanation · 15:00–19:59

and see what is the least recent research that is happening in this field.

Key details

- As I was telling, basically Euler's theorem is to relate homogeneous function with their partial derivative.

- See, that's what Euler observed.
- But if I have a partial derivatives of a function of two independent variables, how am I going to relate them with the degree of their homogeneity?
- and then it has extensive applications in economics.
- Actual form of oiless theorem they use in production function, marginal productivity, elasticity of demand.
- I didn't put the equations in detail because,, most of you are engineers and you might want an engineering application to Euless theorem.
- And intensive variables are homogeneous functions of degree.
- Maybe in your physics course you can mm better understand what is happening and what is the Euler's theorem.
- what does it say and other things, right?
- Because the Euler's theorem is the most important one wherein you can reduce the problem.
- Actually you don't need to find the partial derivatives at all, right?
- The first thing that we did is we defined a homogeneous function.
- We defined what's a homogeneous function.
- Basically, along with the degree.
- And if the degree of the homogeneous function, then Euler's theorem says that x into ux plus y into u y is equal to n into u.
- And then we also observe the extension of this for the second order.
- N independent variables.
- So in the next session we will solve problems on Euler's theorem.

Example or application from the lecture

- So basically it also gives the scaling of the given function, that's because of n into u, that's what we discussed just now.
- So here I have taken three independent variables and I have given first order.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Homogeneous functions and Euler's theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice